

Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

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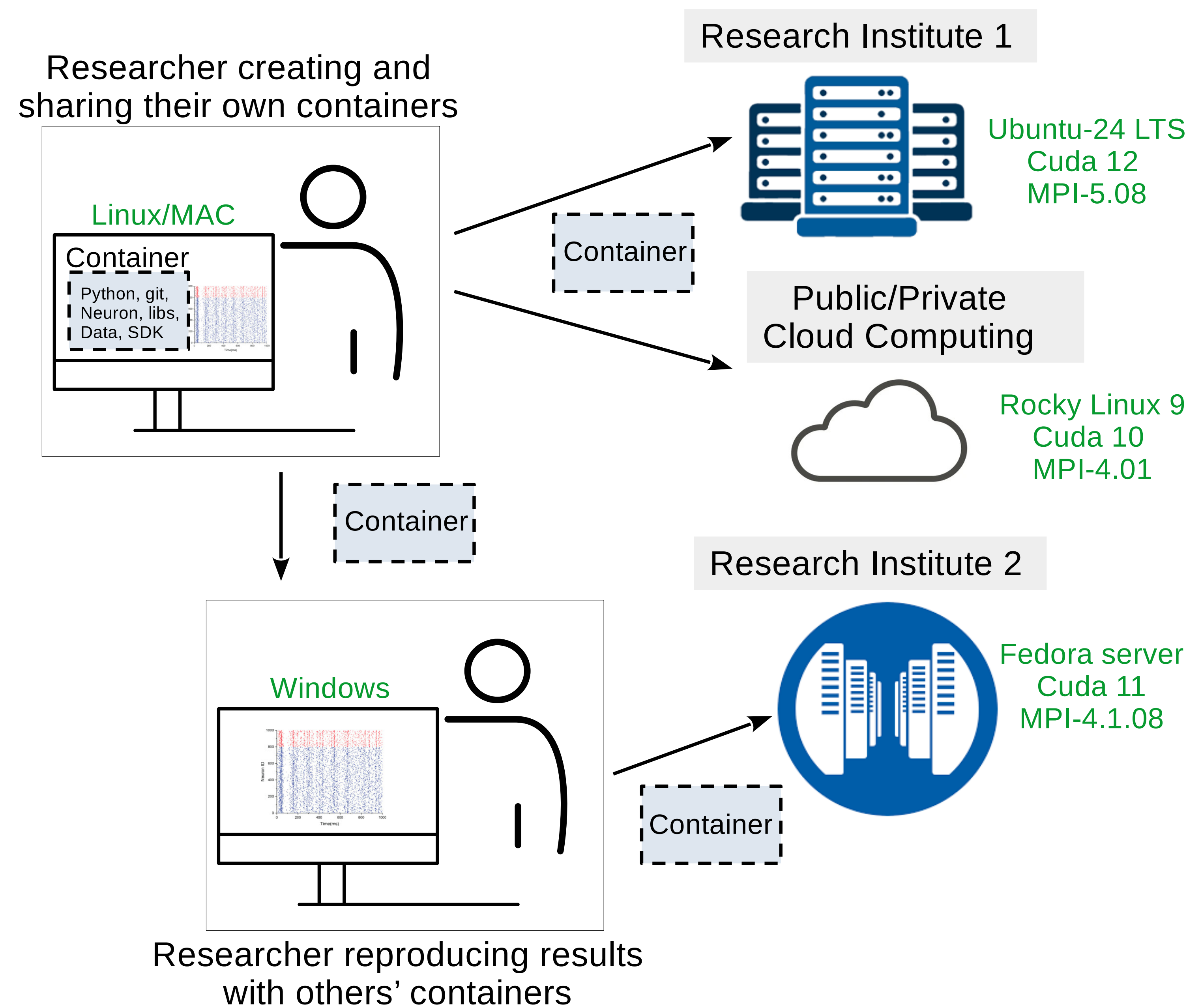
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Introduction

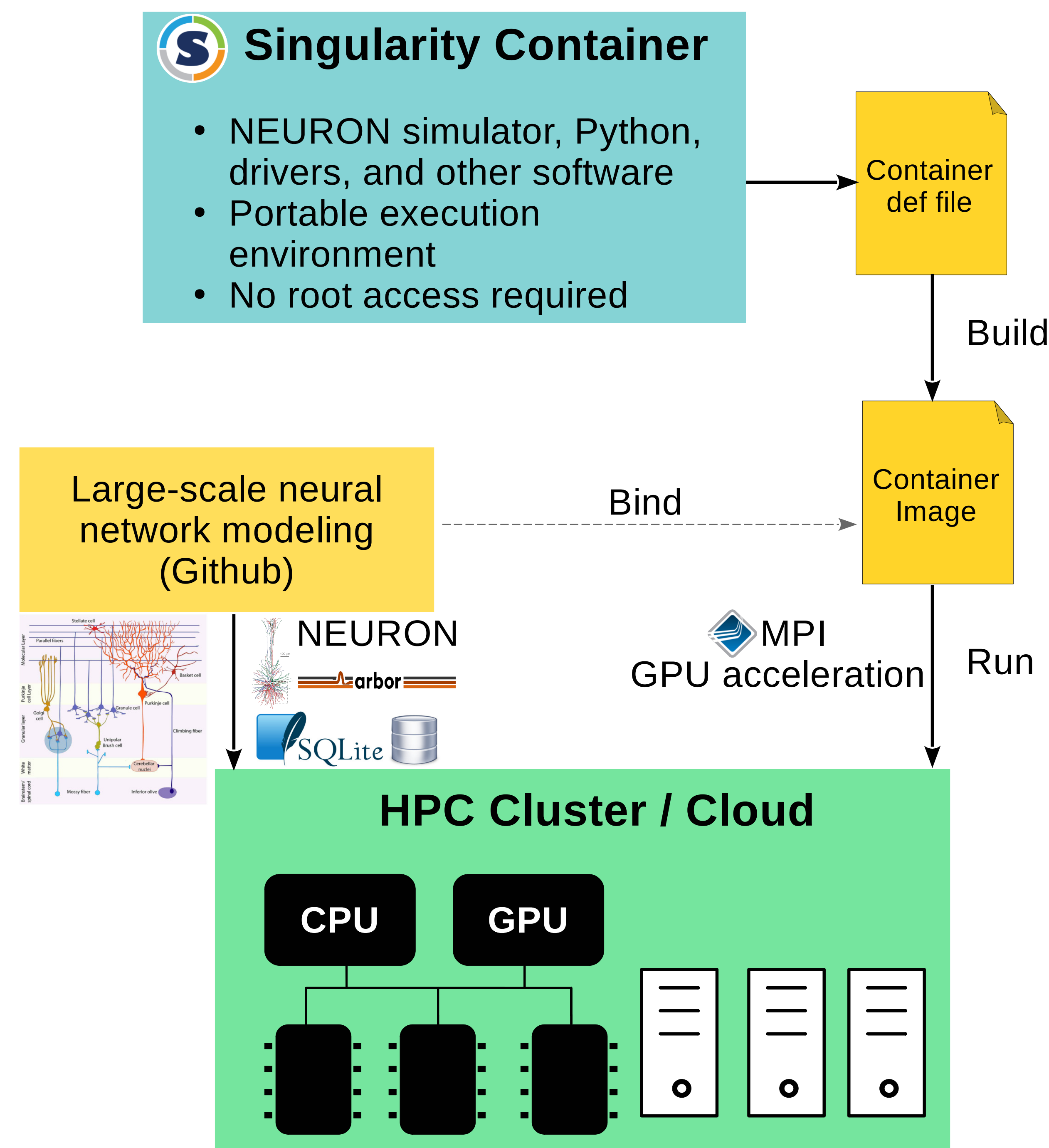
1. Modeling **large-scale, biophysically detailed neural networks** increasingly relies on **high-performance computing (HPC)**.
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to **software incompatibilities** and **complex dependencies**.
3. We address this issue using **Singularity containers** that encapsulates the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.
4. We give practical demonstrations by benchmarking HPC environments with **cerebellar mossy fiber–granule cell network** simulations.

Containerization for Scientific Computing

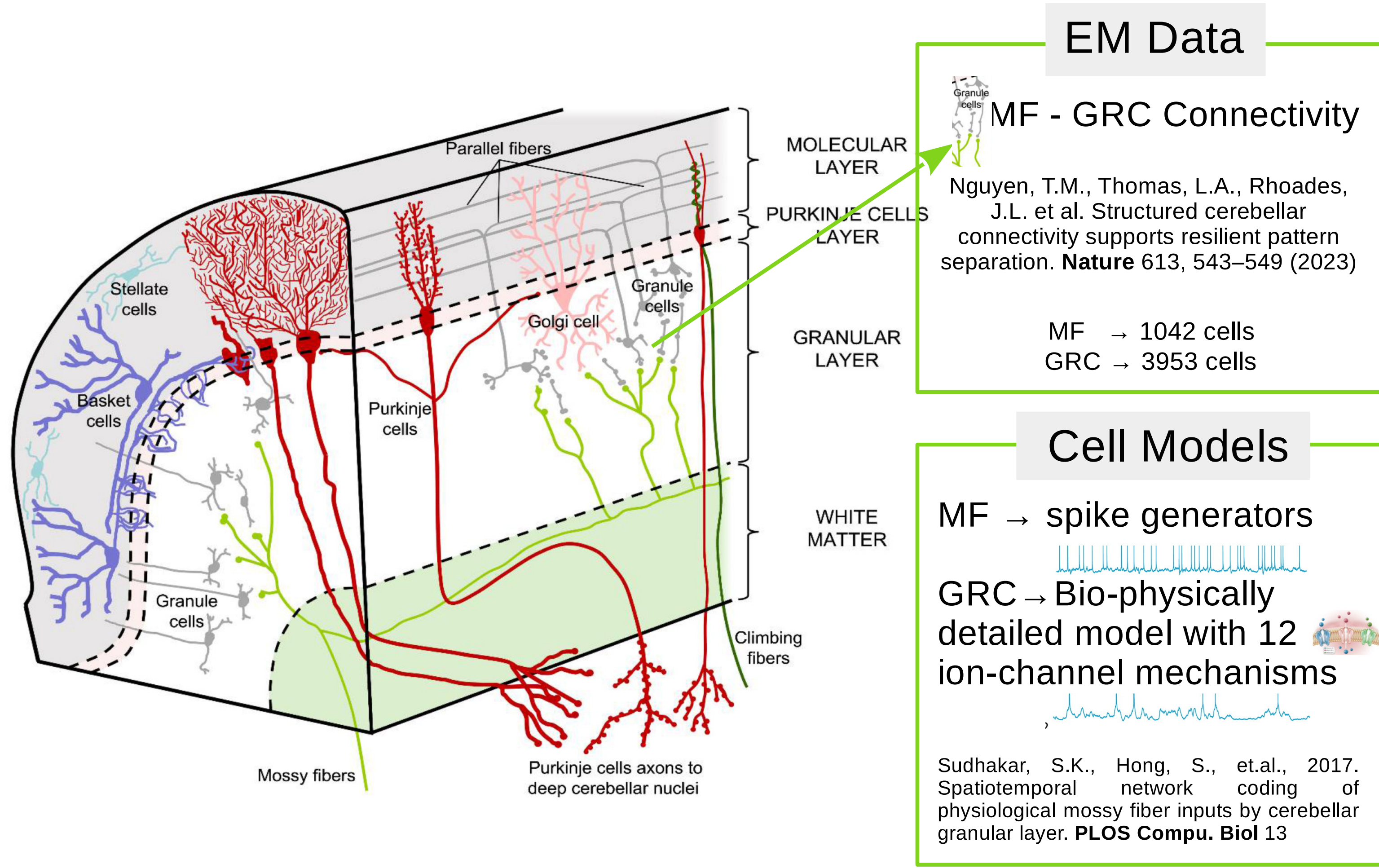
A container is a lightweight, standalone, virtualized executable environment that includes all dependencies required to run large-scale scientific models.



Our Computational Framework



Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



We run multiple copies ($n = 280$) of the network connectivity model simultaneously to simulate extreme computational workloads.

This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

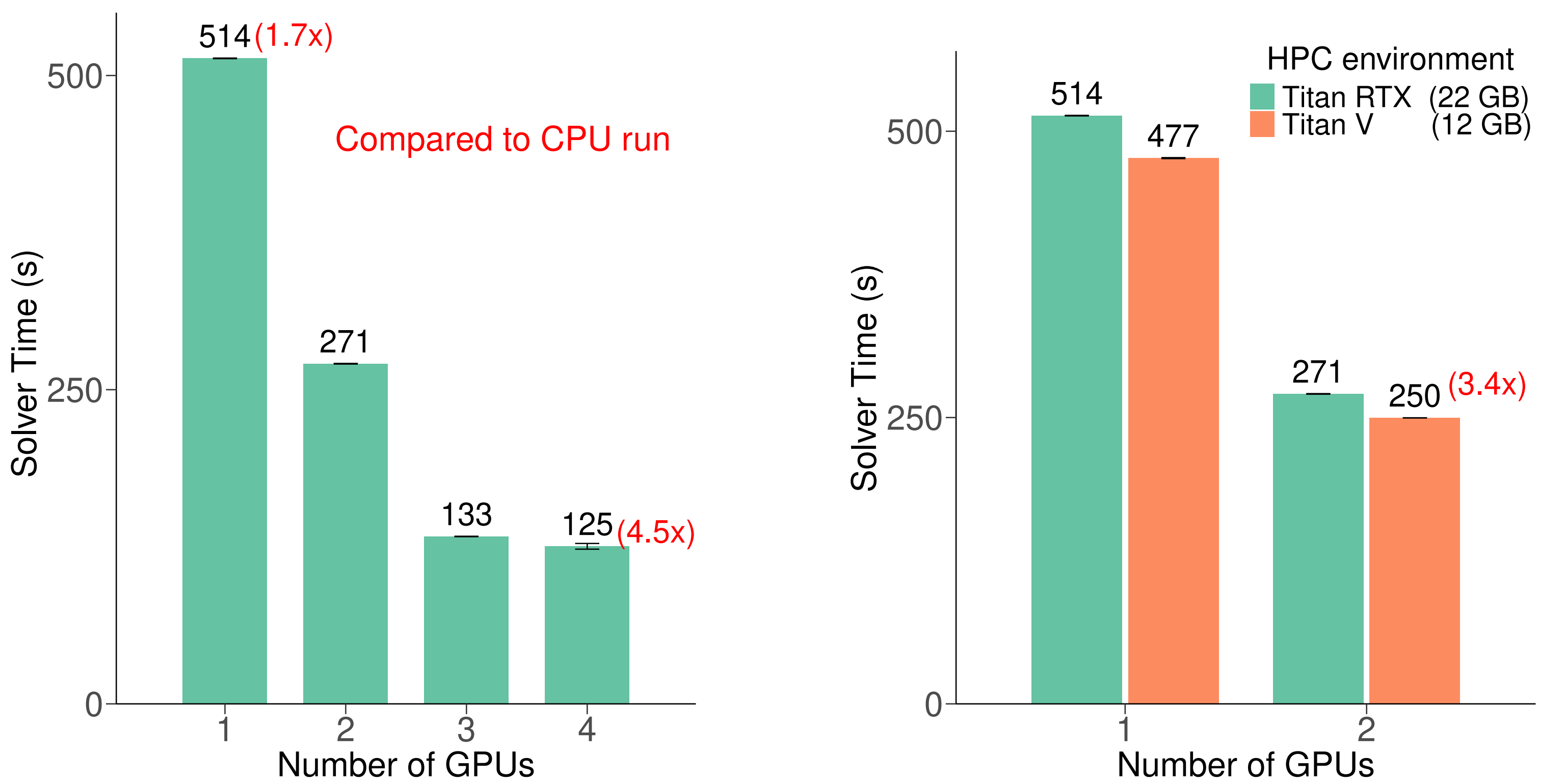
Table: Network Configuration and Simulation Parameters

Number of MF	0.30 M
Number of GRC	1.09 M
Number of compartments	3.29 M
Number of synapses	3.47 M
Simulation time	1 s
Simulation software	CORENeuron

Benchmarking HPC Environments

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	564	CORENeuron-CPU



(a) TITAN RTX architecture: Solver time vs. varying numbers of GPUs in CORENeuron-GPU simulator.

(b) Comparison of solver time between HPC environments

TITAN RTXs outperform 30-core CPUs by up to 4.5x, with performance improving as GPU count increases.

Future works:

🔧 **Develop a biologically realistic cerebellar circuit model** including granule layer (with Golgi cells), Purkinje layer, and other interneurons.

🔧 **Benchmark network performance across multiple institutions** to evaluate and understand the scalability and performance limits of HPC systems.

Acknowledgment

This work was supported by IBS-R001-D2-2025-a004D5

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